

Project Initialization and Planning Phase

Date	30 September 2026
Team ID	PER206AIFITTRACK
Project Title	AI-FitTrack-API
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform workout tracking and fitness coaching using generative AI, boosting consistency and accuracy. It tackles fitness fragmentation and rigid plans, promising personalized workouts, reduced decision fatigue, and data-driven progress. Key features include Google Gemini 1.5 Flash AI recommendation, automated workout CRUD logging, and consistency insights.

Category / Section	Parameter	Technical Specification (AI-FitTrack-API)
Project Overview	Objective	To deliver an AI-augmented backend enabling users to log workouts, track calorie expenditure, and obtain instant, equipment-tailored exercise routines from Google Gemini 1.5 Flash.
Project Overview	Scope	Covers user authentication, JWT route protection, comprehensive workout CRUD, multi-parameter search/filtering, and real-time AI recommendation and insight endpoints.
Problem Statement	Description	Rigid, non-adaptive exercise regimens and disorganized manual logging lead to high dropout rates and workout plateaus, while commercial alternatives impose steep recurring subscription fees.
Problem Statement	Impact	Lack of guidance causes overtraining injuries and fitness stagnation. AI-FitTrack-API provides free, scientifically structured exercise guidance accessible via REST API.
Proposed Solution	Approach	Pairs an asynchronous Node.js and Express backend with MongoDB persistence and Google Gemini 1.5 Flash generative intelligence for sub-50ms CRUD and <1.5s AI responses.
Proposed Solution	Key Features	Stateless JWT auth, Mongoose schema validation, workout CRUD, MET calorie estimation, multi-param search, Gemini workout generator, and AI consistency analytics.
Resource Requirements	Hardware	Local Dev Server: Multi-core CPU, 8 GB RAM, 100 GB SSD storage; Cloud container deployment readiness on AWS EC2 or Google Cloud Run.
Resource Requirements	Software & Tools	Node.js (v20+), Express.js framework, MongoDB (v7.x), Mongoose ODM, jsonwebtoken, bcryptjs, @google/genai SDK, VS Code Thunder Client.